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Engineering Sustainability

Development of a digital twin-driven bridge construction safety monitoring system

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The construction phases of bridges involve a wide range of elements and tasks, generating a vast amount of data in terms of type and volume. Traditional manual and document-based methods are no longer sufficient for effective and accurate project management. Consequently, more comprehensive and systematic management approaches are required. This study proposes a three-dimensional digital twin information display platform that links with the actual state of the structure, facilitating bridge construction monitoring and safety warning through the integration of digital twin technology with multi-source monitoring information. A dynamic multi-resolution digital twin model for bridge construction has been established with a bridge digital model and structural finite element model. Then a structural evaluation and warning method focusing on key stress and deformation indicators has been comprehensively formulated. A comprehensive digital twin information monitoring and management system platform has been developed for bridge construction, complete with a feedback mechanism for warning information. The experimental case application in the Zhanlin Yellow River Bridge construction process demonstrates that this method enables flexible management of the construction process and dynamic evaluation of safety status. These advancements contribute to ensuring bridge construction safety and provide administrators with a digital, real-time online management platform.

Keywords: bridges/construction/digital twin/health & safety/monitoring

1. Introduction

With the continuous development of the socioeconomic, the total volume of road transportation is increasingly being climbed, and the issue of overloading vehicles is becoming more prominent (Gomez and Vassallo, 2015; Li *et al*., 2020; Lv *et al*., 2014). In addition to design and construction defects, improper operation and maintenance, as well as material performance deterioration and many other factors, a large number of bridges in service are in a state of "working with disease," which poses serious safety hazards (Feng *et al*., 2023). The increasingly critical situation of the operational safety of bridges has led the international community to recognize the importance of adopting more systematic and targeted technical measures during the service life of bridges in order to respond more effectively to the growing challenges of bridge safety (Han *et al*., 2018; Jeon *et al*., 2023; Somodikova *et al*., 2016; Zacchei *et al*., 2022; Deng *et al*., 2020). Therefore, the safe service of the existing bridges will be the main direction of bridge engineering in the future, and at the same time, it is urgent to vigorously develop new bridge intelligent operation and maintenance technologies such as intelligent detection, monitoring, evaluation, and maintenance.

At present, researchers at domestic and overseas have established bridge construction monitoring and operational maintenance systems, which have initially realized the digital collection and management of data at various stages. Wu *et al*. (2011) combined GNSS-RTK (global navigation satellite system real-time kinematic) and GIS to develop a set of cable hoisting construction safety monitoring system, which can carry out real-time positioning of cable hoisting tanks with high accuracy and assist command. Cheng (2021) developed a steel bridge construction assembly simulation system based on Unity3D, which is able to guide the process of construction based on the progress information or component location. Sun *et al*. (2020) established an automatic wireless monitoring system by means of a sensor system, data acquisition and communication system, transmission system, and monitoring center. Chen and Zhang (2020) built a multichannel monitoring system of vibrating string strain sensors to solve the problems existing in traditional data collection and transmission. Mei *et al*. (2020) introduced PTDMS (pressure-transmitter-based communicating pipe deflection monitoring system), which is commonly used in bridge health monitoring, into the deflection monitoring of the main girder of Yangzi River Bridge during the construction period from the perspective of whole-life monitoring.

Given this context, digital twin technology emerges as a promising digital solution for bridge structural monitoring and risk analysis. It offers the capability to create a highly integrated environment

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by incorporating advanced sensing and simulation technologies into cohesive solutions (VanDerHorn and Mahadevan, 2021). This integration is applied to analyze, predict, infer, and support decision-making for complex systems. Its feasibility and effectiveness have been demonstrated and validated across various disciplines, yielding significant benefits and development potential (Rasheed *et al*., 2020; Semeraro *et al*., 2021). In the field of bridges, prior studies have explored the application of digital twin technology in structural health monitoring and load analysis. For instance, Ozer and Feng (2019) developed a CPS framework encompassing finite element models of bridges, smartphones, and centralized and distributed computing facilities. Structural vibration information was captured via smartphones and wirelessly transmitted to servers for modal frequency identification, finite element model updates, and structural reliability assessment. Dan *et al.* (2022) established a digital twin monitoring system for full-bridge condition perception and safety warnings by integrating weigh-in-motion (WIM) technology with heterogeneous machine vision data. Kang *et al.* (2021) proposed a multimedia knowledge-based digital twin framework for bridge health monitoring, synchronizing physical and virtual spaces by using sensor-acquired data to update simulation models and simulate extreme scenarios to ensure structural integrity. However, these studies primarily focus on the operational phase of bridges, emphasizing monitoring and inspection. Limited attention has been paid to applying digital twin technology to ensure structural safety during the construction phase, leaving room for further exploration and development in this area.

In addition, in terms of management platform, researchers were focused on realizing the standardization and organization of the work for personnel management, progress coordination, quality inspection, and material management, thus improving the efficiency of the work. Ding *et al*. (2021) adopted BIM technology for efficient safety management of the construction process in the Yellow River bridge. Lin (2018) developed a BIM collaborative management platform based on the cross-sea channel project, which constructed a unified data storage and realized the collaborative sharing both business and users. Daniotti *et al*. (2020) created a coding system based on the concept of standardization, where data will be freely accessible through a web-based portal based on the INNO VANCE database, and where objects can be created, updated and linked directly from the BIM software. Tao *et al*. (2018) proposed an engineering collaborative management platform that enables large-scale, real-time processing of incoming management data, as well as facilitating collaboration, manipulation, and analysis of data throughout the lifecycle of a construction project.

Comprehensively, the above research shows that the research on bridge construction monitoring system and management platforms has initially realized the informatization. Moreover, it enabled a shift in management efforts from manual domination to online management as well as collaborative work. However, different systems are independent from each other, which prevents the effective transfer of data, so that information on individual phases or types of work cannot be shared and managed in multiple systems (Pipinato *et al*., 2019; Sung *et al*., 2016; Zhong *et al.*, 2023). Meanwhile, in the face of the large amount of data collected by the information technology system, the processing, analysis and calculation capabilities are insufficient (Yu *et al*., 2013). As a result, the value of the data is not fully utilized, and it is still a long way from realizing the wisdom of bridge management Accordingly, combined with Zimlin Yellow River Bridge project, this paper integrates multi-source data of the whole life cycle of the bridge and realizes standardized data management based on BIM and digital twin technology. Simultaneously, it builds a three-dimensional display platform of bridge information linked with the actual state of the structure and carries out bridge construction monitoring as well as safety warning based on the fusion of multi-source information. Furthermore, it promotes the development of informatization and intelligence in the bridge industry, while also guaranteeing the structural safety and long-life service of bridges.

2. The research logical framework

The digital twin-based logical framework of the bridge construction process data monitoring and safety evaluation proposed in this study is illustrated in Figure 1. It can be divided into two main

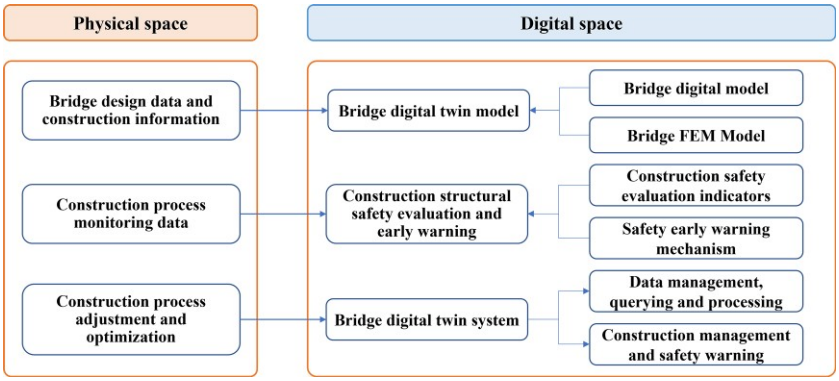


Figure 1. The logical framework for bridge construction data monitoring and safety evaluation based on digital twins

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parts: the physical space and the digital space. The physical space primarily encompasses three key components: the actual design information and construction procedures of the bridge, which provide the foundational data required for constructing the bridge digital twin model; the monitoring data collected during the construction process, which is used for real-time data interaction with the digital twin model to assist in the safety evaluation of the construction process; the adjustment and optimization of the construction process, serving as feedback from the digital twin model.

The digital space consists of three main components: the digital twin model of the bridge, which includes both the digital model of the bridge and the finite element analysis model, serving to simulate the theoretical state of the bridge during the construction process; the construction structural safety assessment and early warning system, which includes evaluation indicators and warning mechanisms to analyze and determine whether there are potential safety hazards in the various stages of the bridge construction process; and the digital twin platform, which provides a user-friendly data management, query, and operation system to feedback and guide the adjustment and optimization of the construction process.

The physical space and digital twin are interconnected through real-time data transmission, enabling data transfer, mapping, and synchronization. First, the construction of the bridge's digital twin model requires reference to the bridge's design data, including its geometry, materials, and so on, and adjustments are made based on the actual completed state of the bridge. The structural state data of the bridge, collected during the construction process in the physical space, is transmitted to the twin platform via wired or wireless networks. Ultimately, the results of the analysis and calculations performed by the twin platform are fed back to the devices of the on-site personnel.

3. Bridge digital twin model construction and simultaneous simulation

3.1 Bridge digital twin model construction

To establish a bridge digital twin system platform, it is essential to construct a digital bridge model to provide virtual digital assets that serve as the central carrier for data integration, presentation, and interaction. This study utilizes BIM and 3D modeling software to initially create the digital bridge model based on fundamental bridge design data. The established model encompasses the main structure of the bridge, deck slabs, piers, abutments, and other components. The modeling logic for the primary bridge structures and components is as follows: (1) Overall modeling sequence: a) bridge deck system; b) superstructure; c) substructure; (2) Modeling sequence of bridge deck system parts: a) bridge deck pavement, expansion joints; b) parapets, crash walls; c) lighting, signs; (3) Modeling sequence of superstructure parts: a) main beam; b) hinge joints, wet joints; c) beams, diaphragms; d) bearings; (4) Modeling sequence of substructure parts: a) cover beams, table caps; b) pier body or columns, platform and other abutment appurtenances; c) pier foundations, bearing platforms and pile foundations.

After completing the creation of all structure and component models, web-BIM model conversion technology is used to convert the bridge and its associated models into a standard 3D model format (e.g. .glb, .gltf) and model asset files (e.g. JSON/XML data files). This process enables the separation of the model from its numerical data and ensures lightweight processing. Based on this, a web-based 3D model rendering framework (e.g. three.js, babyon.js) is utilized to load and display the bridge model on the web. The result of loading the established bridge model on the web is shown in Figures 2 and 3.

3.2 Simulation of bridge structures based on digital twins

Another core component of the bridge's digital twin model is the bridge's twin simulation model. By executing simulations and analyses on a high-fidelity virtual model of the bridge, it is possible to gain insights and analytical results that exceed those obtainable from the physical structure alone, providing highly detailed analysis data. In this study, Midas Civil 2019 was utilized to perform a comprehensive simulation of the entire construction process during the main bridge's incremental launching stage. The basic geometric dimensions of the simulation model were derived

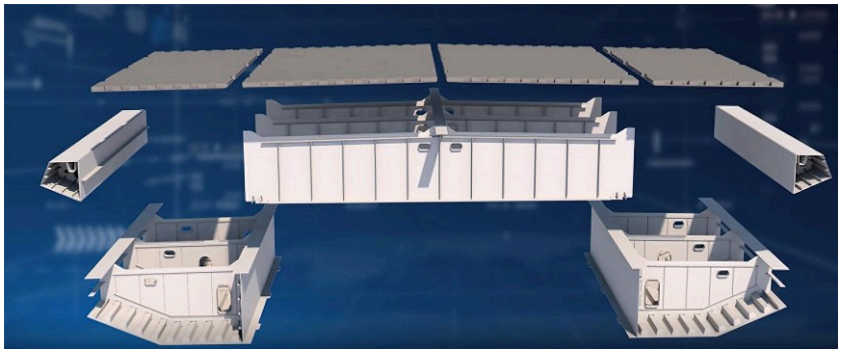


Figure 2. The established model of the bridge deck slab

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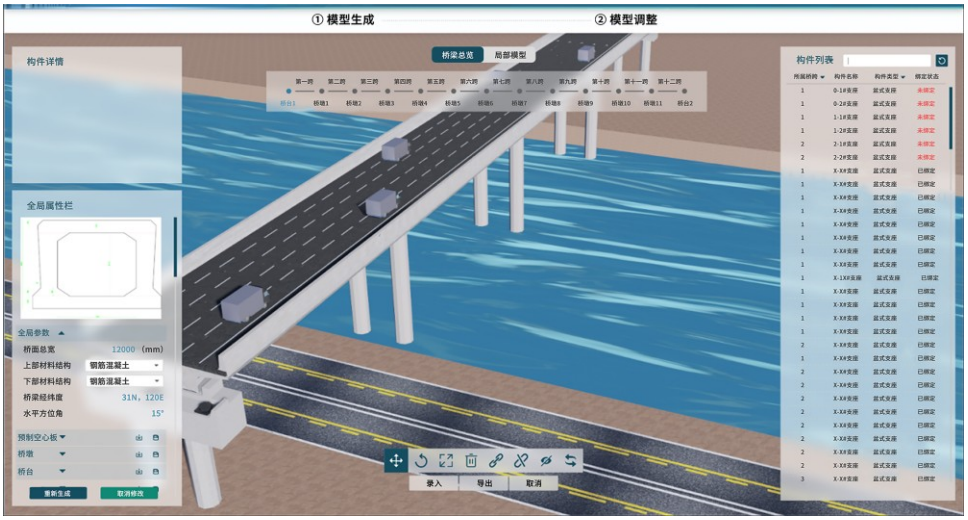


Figure 3. Loading and displaying the built bridge digital model on the web side

from the bridge virtual model. On this basis, the necessary parameters for mechanical simulation analysis were set. The final simulation model, as shown in Figures 4 and 5, includes 267 rod elements, 604 nodes, and 19 construction stages. The model adopts a fixed-beam moving-pier approach, where the position of the beam remains constant while the temporary pier's location is adjusted to simulate different stages of the incremental launching process. The temporary pier is modeled as a compressive-only support with a stiffness of 1×107 kN/m, and the composite beam is simulated using a composite section element. The ultra-high performance concrete (UHPC) has an elastic modulus of 50 GPa and a unit weight of 28.5 kN/m^3, while the steel main beam is

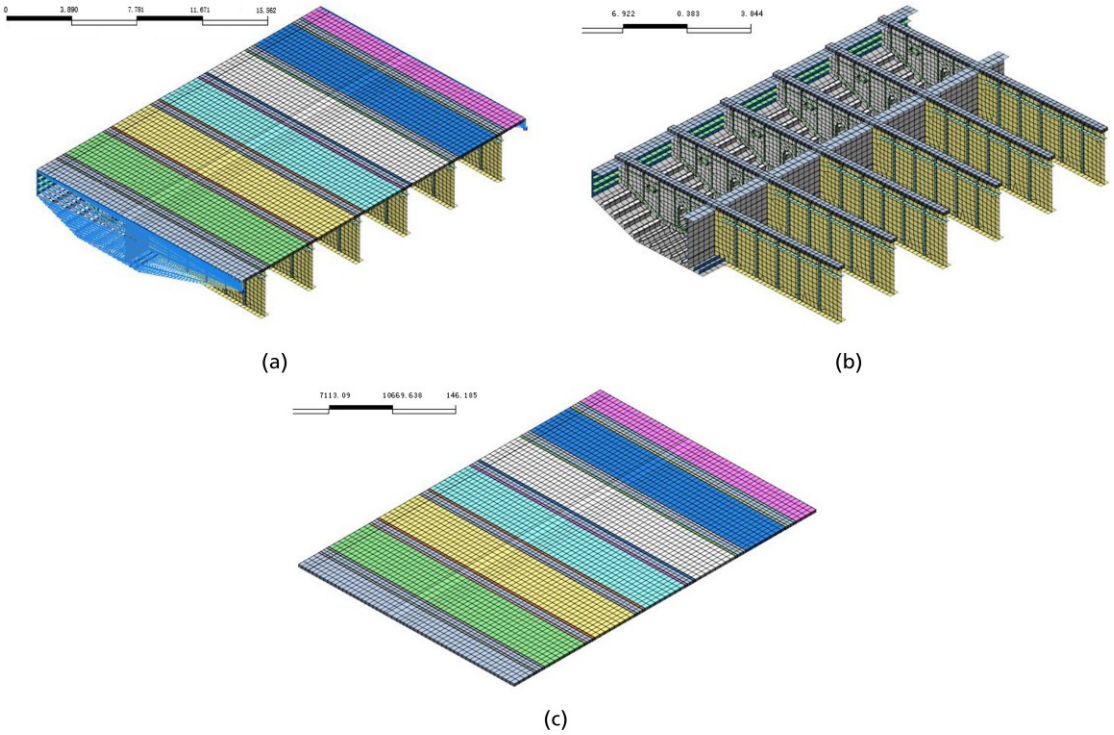


Figure 4. Finite element model mesh for steel-hybrid combination beams: (a) integral mesh of steel-hybrid combination beams; (b) mesh of steel structures; (c) mesh of UHPC bridge deck

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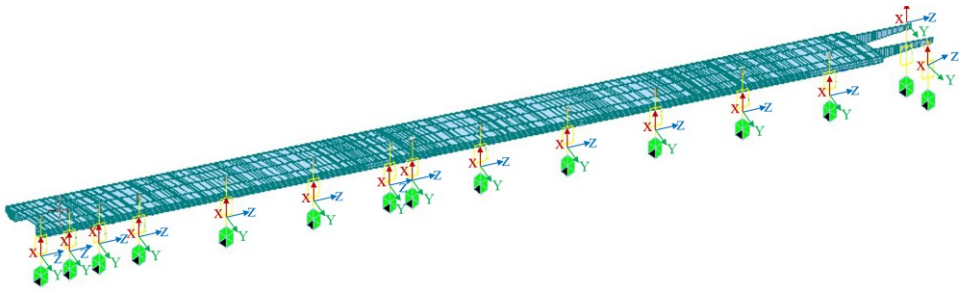


Figure 5. Finite element model for simulation and calculation of jacking process

made of Q345 steel. The primary simulation parameters are listed in Table 1.

4. Construction structure safety analysis and warning method based on digital twins

4.1 Structural safety evaluation indicators

To ensure timely and effective structural safety warnings during bridge construction, representative factors must be selected as early warning indicators. According to bridge construction standards, the constructed bridge structure must possess sufficient strength, stiffness, and stability. Strength and stiffness are typically reflected through indicators such as internal forces and deformation, which are critical points of focus for monitoring during construction. While stability is generally addressed in the design phase with stability checks, in the case of cable-stayed bridge construction, internal forces and deformation indicators should be prioritized for continuous monitoring and evaluation.

4.1.1 Internal force indicators

Internal forces are indicators of the physical and mechanical state of a bridge structure, relating directly to the strength and stability of the structure. In the construction of cable-stayed bridges, internal force indicators include stresses in the main tower and the main girder, with key sections selected for monitoring these parameters.

4.1.2 Deformation indicators

Deformation indicators are crucial for ensuring the safety of cable-stayed bridge construction, reflecting the stiffness and stability of the bridge structure. These indicators primarily include: the deformation of the main girder and the deformation of the main tower. For the main girder, deformation must be measured at each construction stage and under each working condition to ensure that the girder's elevation remains within controlled limits. The deformation of the main tower involves measuring the cross-sectional dimensions during the construction phase to ensure that the tower's profile remains smooth and within allowable limits.

After identifying the warning indicators, standardization of these indicators is necessary to facilitate the analysis and assessment of the bridge's safety status. In this study, stress and deformation are selected as the primary warning indicators. The stress indicator is processed by calculating the percentage difference between the theoretical and measured values, while the deformation indicator is handled by determining the difference between the theoretical and measured values.

4.2 Safety early warning mechanism during construction

The reasonableness of the design of warning criteria is crucial to the quality and reliability of the warning system. This study, in conjunction with bridge construction control standards, proposes a set of warning criteria that integrates mechanical principles, computational models, and relevant bridge construction standards. The criteria are introduced as follows:

4.2.1 Warning criteria for stress indicators

- Criterion A: Measured stress values that are less than or equal to the allowable stress as specified by the standards indicate that the bridge structure meets the regulatory requirements; otherwise, it is considered abnormal.
- Criterion B: For prestressed concrete components that are fully prestressed, the cross-section should not exhibit tensile stress; otherwise, it is considered abnormal.

Table 1. Materials and parameters of structural elements in the jacking phase of the main bridge

Number	Structure	Material	Modulus: MPa	Bulk density: kN/m^3	Strength: MPa	Note
1	Bridge deck	Ultra-high performance concrete (UHPC)	50 000	28.5	Tensile strength: 20 MPa; Compressive strength: 120 MPa	Uniformly distributed load
2	Steel box girder	Q345qD	206 000	/	345 MPa	Uniformly distributed load
3	Guide beam	Q345qD	206 000	/	345 MPa	Uniformly distributed load

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- Criterion C: Using the theoretical stress of the tower and beam as a reference, if the measured stress deviates by more than 20% above or below the theoretical stress, the data is considered abnormal.

4.2.2 Warning criteria for deformation indicators

- Criterion A: In two consecutive construction conditions, the trend of change between the measured values and theoretical values should be consistent; if they diverge, it is considered an abnormal situation.
- Criterion B: The deviation of the pylon and main girder axis should be less than 10 mm; otherwise, it is considered abnormal.
- Criterion C: The deviation of the pylon cross-sectional dimensions should be less than 20 mm; otherwise, it is considered abnormal.
- Criterion D: The inclination of the construction tower column should not exceed 1/3000 of the tower height, and should be within 30 mm or the design requirements; otherwise, it is considered abnormal.
- Criterion E: The elevation error of the beam segment (compared to the calculation model) must be within the allowable range of ± 20 mm (for $L \leq 100$ m) or $\pm L/5000$ (for $L > 100$ m), where L represents the bridge span. Values exceeding this range are considered abnormal.

The warning indicators are used to assess the system's safety state based on how far the indicator values deviate from theoretical values. Here, we establish a Cartesian coordinate system, as shown in Figure 6. The warning indicator is set as X, while Y represents the trend of danger development, with O indicating the theoretical value and the measured values lying within the X-axis range. The measured value within the range (a, b) represents the safety zone, while (c, a) and (b, d) denote the precautionary safety zone. The ranges (e, c) and (d, f) are classified as dangerous zones, and ($-\infty$, e) and (f, $+\infty$) are considered extremely dangerous zones. In practical engineering, based on the aforementioned method of defining warning zones and combined with warning criteria, the warning thresholds for the actual construction of the cable-stayed bridge are determined according to the bridge's specific conditions, as shown in Tables 2 and 3. Table 2 illustrates the deformation warning thresholds and levels, encompassing section size deviation, axis offset, and deformation deviation for the main tower, as well as elevation difference and axis offset for the main girder. Taking the main tower's section size deviation Y as an example, when the absolute value $|Y|$ is less than 10 mm, no warning is issued. For $|Y|$ in the range of 10–20 mm, a Level 1 warning is triggered; in the range of $|Y|$, a Level 2 warning is reached; and for $|Y| > 30$ mm, a Level 3 warning is issued. Similar thresholds and levels apply to the other deformation indicators. Table 3 outlines the stress warning thresholds and levels, including stress in the main tower, main girder, and stay cables. For instance, in the case of the main tower's stress indicator σ, no warning is issued if the absolute deviation between the measured stress and theoretical stress $|\Delta\sigma|$ is less than 10%. When $|\Delta\sigma|$ falls within the range of 10%–20%, a Level 1 warning is triggered; within 20%–30%, a Level 2 warning is issued; and for $|\Delta\sigma|$ exceeding 30%, a Level 3 warning is reached. Similar thresholds and levels are applicable to other stress indicators.

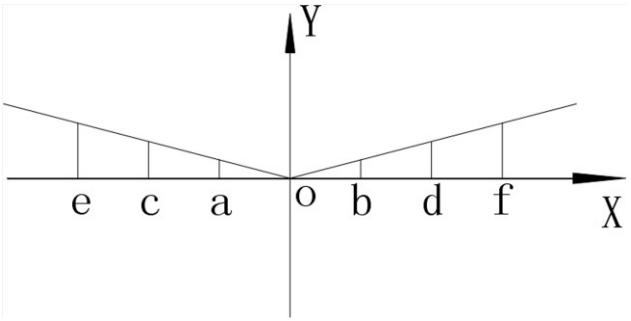


Figure 6. Classification of indicators into ranges with different levels of early warning

5. Bridge digital twin system platform development

5.1 System logical framework

The platform consists of four parts: data acquisition layer, data management layer, calculation and analysis layer, as well as application management layer, whose hardware and software components are shown in Figure 7. The data acquisition layer is the first part of the framework. It mainly contains various types of sensing terminals that acquire structural and environmental information, such as cameras, radars, laser scanners and other automated inspection equipment. After the data is collected by the above devices, edge computing is used to carry out the preliminary cleaning and basic organization of the data by the local servers that come with or are connected to these devices. The processed data is then sent to the central server through the wireless sensor network.

Table 2. Warning ranges and levels of deformation indicators

Structure parts		No warning	Level 1	Level 2	Level 3
Main tower	Section size (mm)	$\|Y\| \leq 10$	$10 < \|Y\| \leq 20$	$20 < \|Y\| \leq 30$	$\|Y\| > 30$
1.	Axis deviation (mm)	$\|Z_s\| \leq 5$	$5 < \|Z_s\| \leq 10$	$10 < \|Z_s\| \leq 15$	$\|Z_s\| > 15$
2.	Deformation deviation (mm)	$\|\Delta L\| \leq 10$	$10 < \|\Delta L\| \leq 20$	$20 < \|\Delta L\| \leq 30$	$\|\Delta L\| > 30$
Main girder	Deflection elevation (mm)	$\|X\| \leq 10$	$10 < \|X\| \leq 20$	$20 < \|X\| \leq 30$	$\|X\| > 30$
3.	Axis deviation (mm)	$\|Z_s\| \leq 5$	$5 < \|Z_s\| \leq 10$	$10 < \|Z_s\| \leq 15$	$\|Z_s\| > 15$

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Table 3. Warning ranges and levels of stress indicators

Structure parts	No warning	Level 1	Level 2	Level 3
Main tower				
Stress	$\|\Delta\sigma\| \leq 10\%$	$10\% < \|\Delta\sigma\| \leq 20\%$	$20\% < \|\Delta\sigma\| \leq 30\%$	$\|\Delta\sigma\| > 30\%$
Main girder				
Stress	$\|\Delta\sigma\| \leq 10\%$	$10\% < \|\Delta\sigma\| \leq 20\%$	$20\% < \|\Delta\sigma\| \leq 30\%$	$\|\Delta\sigma\| > 30\%$
Cable-stayed				
Cable tension	$\|\Delta T\| \leq 5\%$	$5\% < \|\Delta T\| \leq 10\%$	$10\% < \|\Delta T\| \leq 15\%$	$\|\Delta T\| > 15\%$

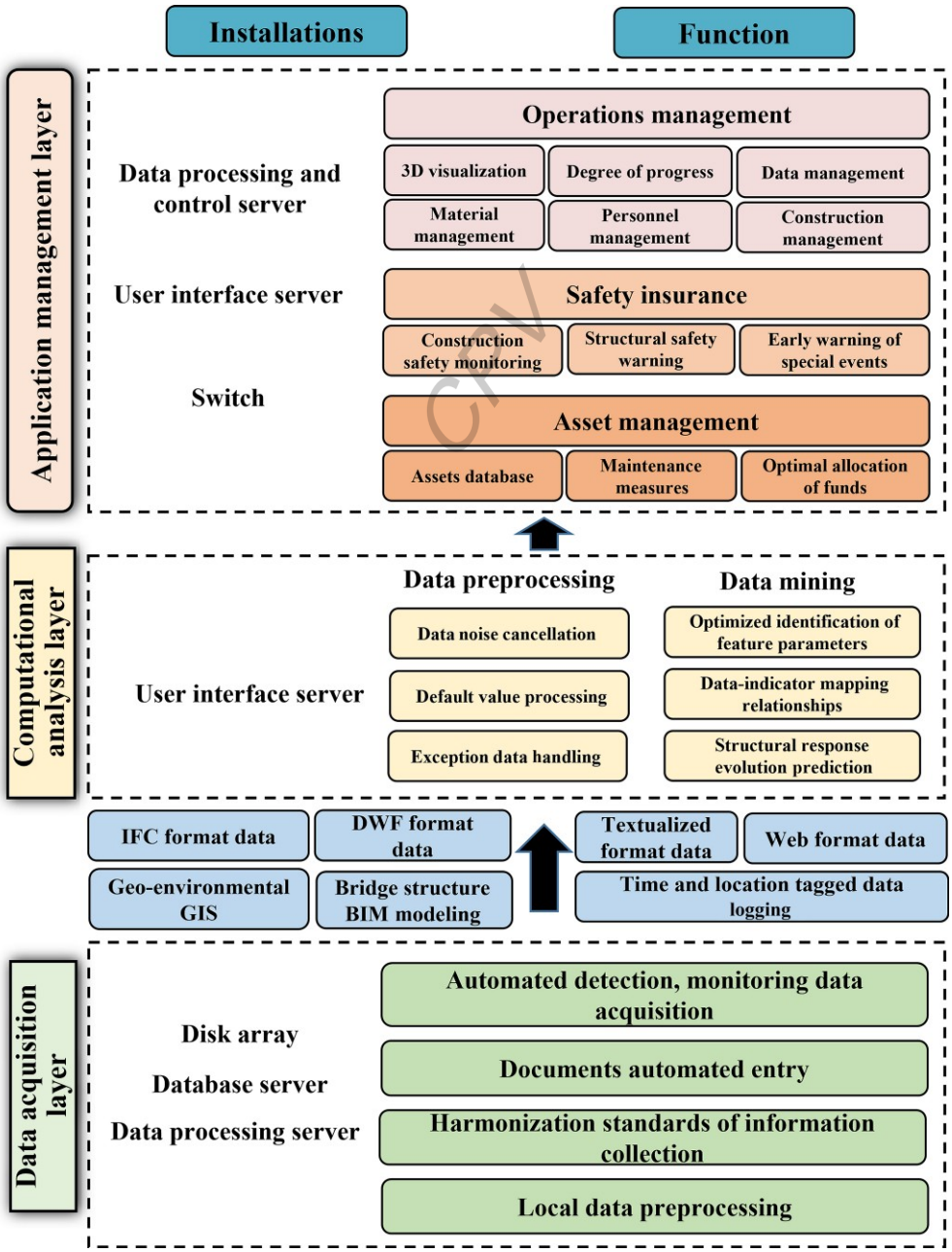


Figure 7. Schematic diagram of platform framework

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Then, next is the data management layer, whose main purpose is to enable integrated management of the acquired data. For the data type dimension, specify the description format and storage format for a variety of data types, and solve the problem of automatic entry and standardization of multi-source data by establishing a unified library of standardized description templates. In the data type dimension, it is able to specify the description format and storage format for different types of data, and solve the problem of automatic entry and standardization of multi-source data by establishing a unified library of standardized description templates. As to the data attribution dimension, the data is bound to its acquisition time and components to realize the orderly storage and retrieval of data.

The computational analysis layer primarily conducts data cleansing and preprocessing to enhance the data quality utilized for calculations (Fan *et al*., 2014; Liu *et al*., 2018; Zhang *et al*., 2017), while also forecasting the future performance of the structure based on data trends. Alternatively, leveraging digital twin technology enables the refinement of the finite element model through the utilization of statistical or raw data (Corral-Acero *et al*., 2020). The finite element model, enhanced by incorporating real data, can more accurately depict the actual changes in the structural state, thereby enabling more precise calculations of the mechanical properties of the bridge structure.

Finally, the platform interface is designed within the application management layer, enabling targeted access to various management application terminals. The management terminal modules are mutually independent and do not interfere with each other. Managers can flexibly combine functional modules based on actual needs, such as monitoring and early warning of structural construction, evaluation of inspection results, and three-dimensional dynamic display of structures.

5.2 System framework

This system adopts a service-oriented architecture based on the B/S architecture framework, utilizing JAVA programming language and incorporating mobile terminal technology. The platform's overall architecture follows a three-tier system, comprising the decision analysis layer, integrated supervision layer, and field application layer in a top-down manner. The comprehensive framework diagram is presented in Figure 8.

5.2.1 Decision analysis layer

The decision analysis layer is built upon the dataset of integrated supervision and field application, enabling comprehensive data analysis across multiple dimensions including objectives, safety, risks, schedule, and resources. Meanwhile, the platform generates decision analysis content in accordance with relevant technical requirements and means through automated processes. The current project's operation status, security status, and progress information can be directly queried through this layer, which facilitates the provision of a solid decision-making foundation for project managers.

5.2.2 Integrated supervisory layer

The integrated supervisory layer aims to establish a mature data standard and uniform data interface for the project. The data information is structured based on various business modules within the application layer, and it is uploaded through the pre-defined interface before being aggregated into the supervision layer. Simultaneously, serving as the intermediary layer, this level also transmits data information to the decision analysis level through a standardized data interface.

5.2.3 Field application layer

The field application layer primarily serves as the central hub for business management and control at construction sites. As the sole module responsible for collecting field data, this layer must ensure

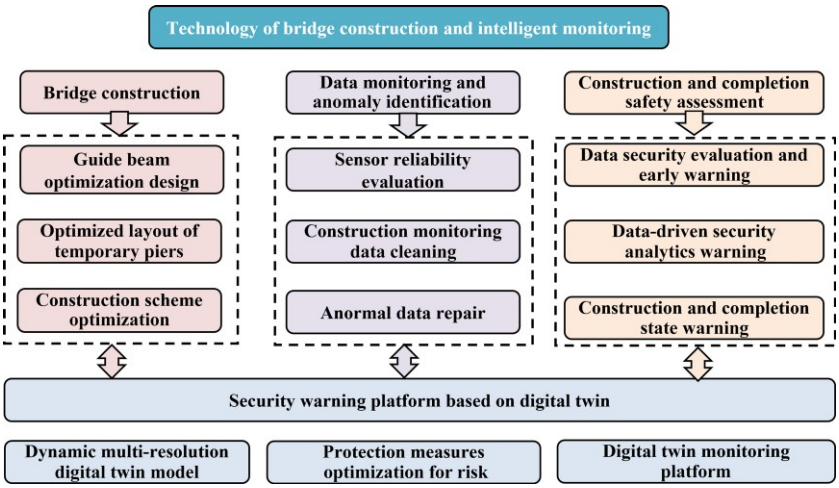


Figure 8. Overall platform framework

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seamless data upload and system-wide management of the entire project. Additionally, apart from real-time data collection, it is imperative to establish a unified gateway and interface to facilitate real-time data upload.

5.3 Functional module design

The platform is primarily designed to effectively manage, visually present, comprehensively analyze, and accurately calculate the data and structure during the bridge construction phase. Digital twin technology is used to connect different objects in different periods, virtual data and physical structures, so as to meet the series requirements of digital management, visual display, comprehensive utilization of data, and structural assessment and early warning, as shown in Figure 9. The main structure as well as the components of the "Human-Machine-Materials-Methods-Environment" are subject to real-time monitoring, enabling the management of structural safety, project progress, and equipment status (Bao *et al*., 2018; He *et al*., 2017; Wang, Li and Hu, 2011). By integrating comprehensive monitoring information, inspection data, specialized testing results, environmental data, and other multi-source datasets, it achieves real-time three-dimensional visualization of bridge structures, surrounding environments, components, mechanical and electrical equipment, personnel activities, materials usage and other elements, and enables early warning for structural safety concerns while facilitating asset management of structures and equipment.

In the aforementioned process, BIM technology relies on the digital model throughout the entire cycle to seamlessly integrate information from various stages, ensuring comprehensive data preservation. Simultaneously, it possesses the temporal label associated with a specific stage and the spatial label linked to a particular component, thereby facilitating vertical data interoperability. The digital twin technology ensures real-time feedback of the actual state of the solid model within the platform, transforming the BIM model from a mere isolated display tool into an integrated solution that facilitates more accurate structural analysis and potential feedback control.

6. Case studies and results analysis

6.1 Case project introduction

This study uses the construction monitoring of the Zhanhua to Linzi Highway Yellow River Bridge in China as a case project. The bridge has a total length of 4630 m. The main bridge consists of a cable-stayed steel-mixed composite beam with two towers and a double cable plane. The main bridge uses the pushing construction method, which poses significant challenges, necessitating comprehensive construction monitoring. The monitoring process entails overseeing a multitude of parameters and a substantial volume of data. However, conventional methods for construction monitoring fail to ensure real-time data acquisition, continuity, and timeliness. Using the digital twin model for construction process monitoring facilitates the visualization of on-site conditions. In the established bridge digital twin system, the digital twin model of the bridge is provided along with visual representations of the project progress and beam segment classifications, as shown in Figures 10 and 11. Figure 10 illustrates the interface and digital twin model of the established bridge digital twin system, showcasing its interactive functionalities and real-time data integration. In Figure 11, the varying colors of the bridge segments in the model represent different structural types, providing a clear visualization of the structural composition and facilitating the interpretation of model-specific information.

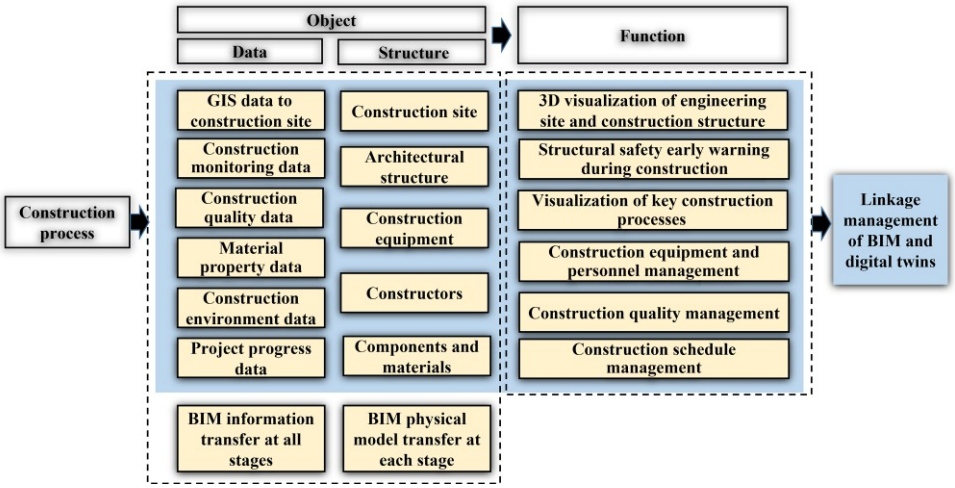


Figure 9. Platform function design drawing

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Figure 10. Bridge digital model representation in an established digital twin system



Figure 11. Classification and presentation of different beam section types of bridges

6.2 Structural data acquisition and monitoring of on-site bridge construction

During the construction process, real-time monitoring of the tower, beam, cables, and temporary structures is conducted to collect monitoring parameters. Based on the monitoring results, necessary adjustments are made to the designed construction process to ensure that the geometric and stress states of the bridge, once completed, closely align with the target design states. A comprehensive digital twin system also includes monitoring of the main tower stress, temporary pier support reactions, temperature data, and linear displacement. To simplify the scope of this paper, the case study focuses primarily on monitoring the main beam stress, introducing the data collection equipment used in the actual project.

The main beam stress (strain) testing sections are selected at stress control sections during construction and at the most unfavourable sections under live load once the bridge is completed. The arrangement of measurement points on the main beam sections emphasizes the strain values at the upper and lower edges. All strain gauges are equipped with reliable calibration data, and the

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corresponding data loggers are used for data collection. A total of 12 stress monitoring sections are set on the main beam (including the midspan of the side span, auxiliary pier support point, midspan of secondary side span, near the main tower base, 1/4 span, midspan, and 3/4 span), as shown in Figure12.

Each section of the steel beam bottom plate of the main beam is equipped with four high-precision surface-mounted steel strain gauges, while four additional high-precision steel strain gauges are installed beneath the UHPC high-performance concrete top plate. Two strain gauges are placed at wet joints, resulting in 10 strain gauges per stress monitoring section and a total of 120 embedded steel strain gauges. The layout of the stress monitoring sections on the main beam is illustrated in Figure 13. The sensor data is transmitted wirelessly via a local area network to nearby monitoring terminal devices. The terminals collect data in real time and perform basic edge computing and processing. The processed data is then transmitted to the bridge's digital twin system through wireless communication networks and mobile networks for real-time status visualization, as well as structural safety analysis and assessment.

6.3 Construction safety data real-time monitoring in digital twins

The structural safety monitoring data from the project site is transmitted in real time to the bridge's digital twin model. Within the digital twin system, project progress can be visually represented using color-coded annotations (as shown in Figure 7), and animations are used to demonstrate the ongoing construction process at each stage. By integrating real-time construction data with the construction model, the bridge model is automatically updated based on jack displacement, significantly enhancing the model's automation and usability. This also enables real-time monitoring of the bridge construction progress. For example, Figure 14 illustrates the construction progress and process of the bridge segments, highlighting the assembled segments and those yet to be completed. Figure 15 presents the critical construction procedures and techniques, serving as a guide for on-site operations.

Simultaneously, during bridge construction, the load-bearing system undergoes rapid changes, accompanied by numerous temporary structures and construction equipment. As a result, the structural stress conditions become complex and variable, necessitating close monitoring for safety. Structural data collected by

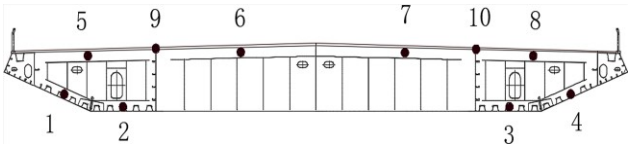


Figure 13. Arrangement of stress measurement points of main girder section

stress and temperature sensors installed on the main girder, main tower, and temporary piers during construction is synchronized and transmitted to the bridge's digital twin system. Within the system's information display and viewing interface, all sensor devices deployed at the construction site can be viewed intuitively. These sensor models are also integrated with real-time monitoring data, allowing users to check sensor locations and detailed information at any time via the visual interface (as shown in Figure 16). The platform records and stores real-time monitoring data, providing online data query, analysis, and export functions, making data access convenient for users. Figure 17 displays the real-time data of support reactions at various temporary piers, while Figure 18 shows the axial, vertical, and horizontal displacement data at various monitoring points during the construction process.

6.4 Structural construction safety evaluation and warning

Structural safety assessment is conducted through preliminary finite element calculations to establish threshold values for various monitoring indicators, ensuring structural integrity. Any values exceeding these thresholds will trigger early warnings. Based on the bridge simulation model discussed in Section 3.2, this case study primarily analyzes the construction safety data of the bridge's main girder structure to demonstrate the implementation of structural safety assessment and early warning within the digital twin system.

Figure 19 presents the stress analysis contour plots of the steel structure of the main girder and the deck panels during the construction process, obtained from the finite element analysis. Each jacking cycle is treated as a computational unit, with the maximum tensile and compressive stresses of the UHPC deck panels and the steel box girder calculated for each cycle. The variation in stress of the UHPC deck panels and steel box girder with the jacking distance is shown in Figure 20 and Table 4. The analysis indicates

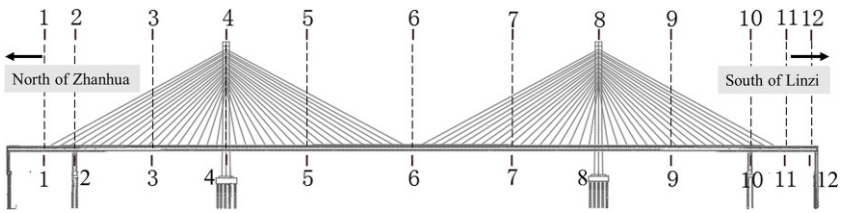


Figure 12. Schematic layout of the stress monitoring section of the main girder of the bridge

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Figure 14. Bridge construction progress monitoring in digital twin system

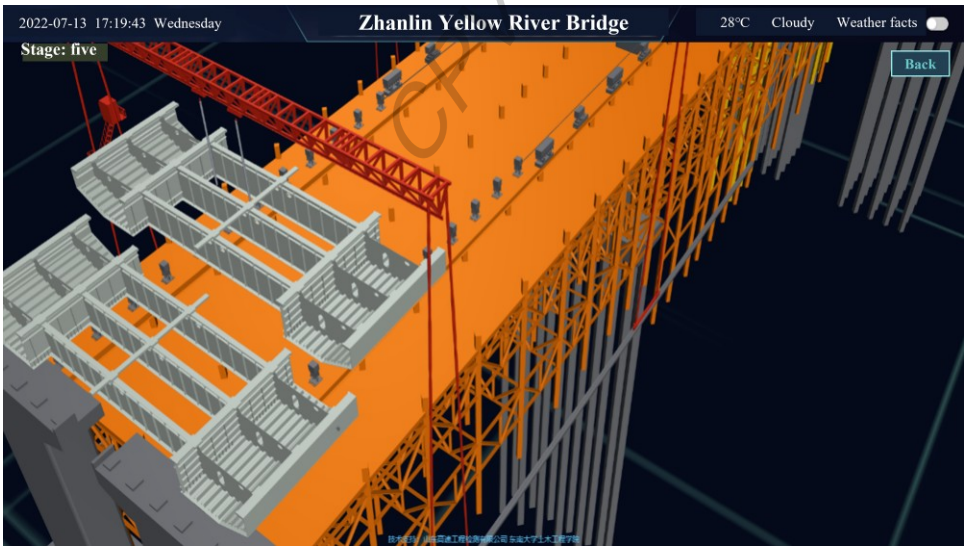


Figure 15. Online display of construction technology integrated with real-time progress monitoring

that although the stress state of the main girder continuously changes throughout the jacking process, the maximum stress variation is relatively small. The maximum stress in the steel box girder is approximately 40 MPa, while the maximum compressive stress in the UHPC deck panel is about 6.63 MPa, and the maximum tensile stress is around 5.07 MPa. All of these values are significantly lower than the material strength limits, indicating that the original construction plan can ensure structural safety.

During the actual construction process, two key indicators were selected for comparison: the main girder stress and the support reaction forces of the temporary piers. The actual jacking distance on-site is approximately 573 m. To date, stress measurement points have been arranged at seven sections, with their distribution locations shown in Figure 21 and detailed in Table 5. In this case study, the measured data from sections , , and are selected for comparison with the finite element analysis results.

The comparison between the calculated stress at the corresponding sections in the finite element model during the jacking of the main girder at 573 meters and the measured results is presented in Tables 5 and 6. The data in the tables reveal that Section , located at the leading edge of the main girder during the jacking process, shows discrepancies between the actual measurements

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Figure 16. Sensor location and status information query and display

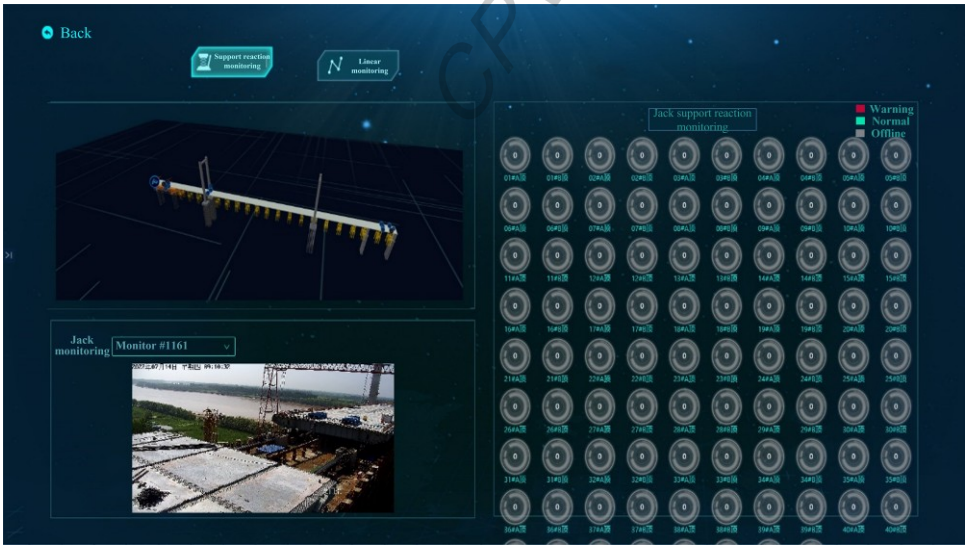


Figure 17. The real-time monitoring of support reactions data at various temporary piers

and the simulation data. These discrepancies are attributed to the influence of the actual construction process, which does not trigger an alert. At Sections and , the measured stress data of the steel box girder and the bridge deck exhibit some deviations from the theoretical values. Specifically, the stress deviation in the steel box girder at Section is 11.04%, which, according to the safety warning mechanism outlined in Section 4.2, triggers a Level 1 warning. The stress deviation in the UHPC bridge deck at this section is 20.40%, resulting in a Level 2 warning. Similarly, at Section , the stress deviation in the steel box girder is 7.04%, which does not trigger a warning, while the UHPC bridge deck shows a stress deviation of 29.70%, leading to a Level 2 warning.

In the bridge digital twin system, the monitoring and alarm module displays alerts in a tabular format, as shown in Figure22, including the following details: structure name, data type, function type, sensor name, alarm level, alarm time, monitoring curves, alarm values, and alarm status. Based on the different alarm levels, relevant personnel receive timely warning information, and the on-site audible and visual alarm systems also respond accordingly.

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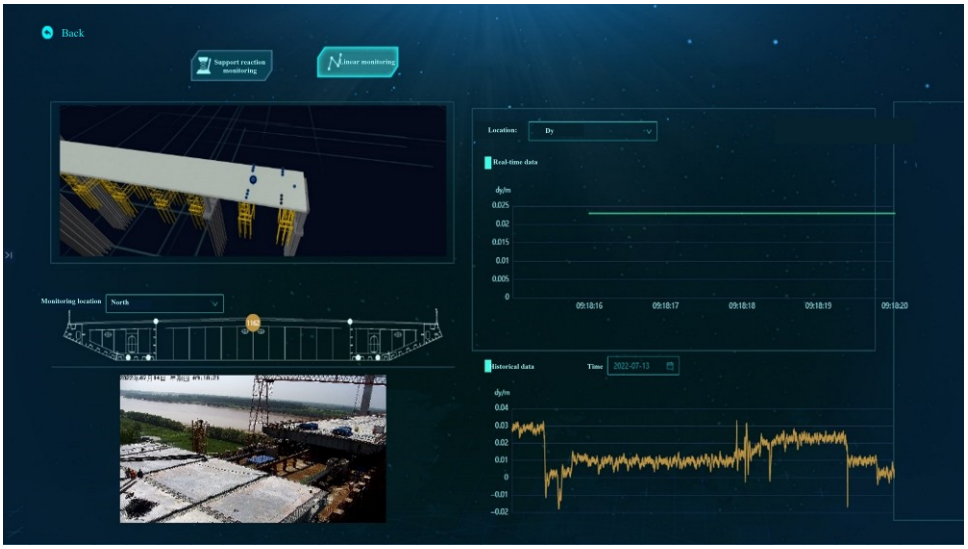


Figure 18. Real-time monitoring of positional offset data

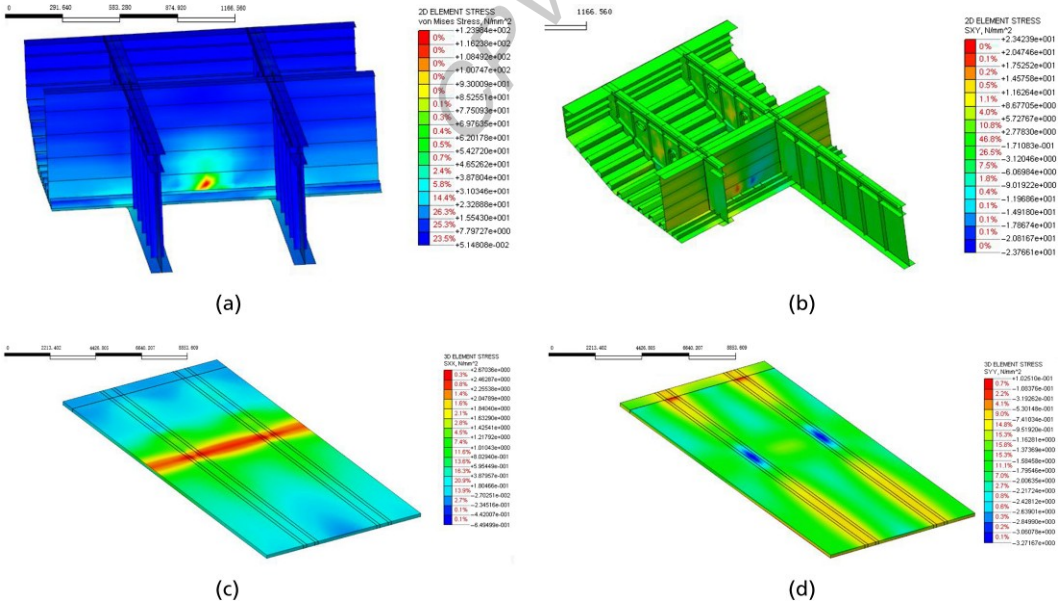


Figure 19. Cloud diagram of local stress calculation for steel box beams and UHPC bridge deck slabs: (a) Mises stresses in steel box; (b) vertical shear stress of steel box; (c) UHPC bridge deck transverse stresses; (d) UHPC bridge deck longitudinal stresses

Upon receiving the warning information, personnel follow the predefined procedures set by the system to address the alerts.

7. Discussion and conclusion

This study investigates a structural safety early warning assessment and monitoring system for bridge construction processes based on digital twin technology. Leveraging digital twin technology and considering the specific requirements of the Zhanlin Yellow River Bridge project, this research establishes a three-dimensional digital twin monitoring platform for the bridge. By integrating multi-source information, the platform links bridge data with its actual state, facilitating construction monitoring and safety early warnings. Built on the digital twin platform, this research ensures seamless integration of data management, safety assurance, and operational control throughout the bridge construction process. It promotes the digitalization of activities such as construction, inspection, evaluation, and maintenance, harnessing the potential of big data to enhance the informatization and

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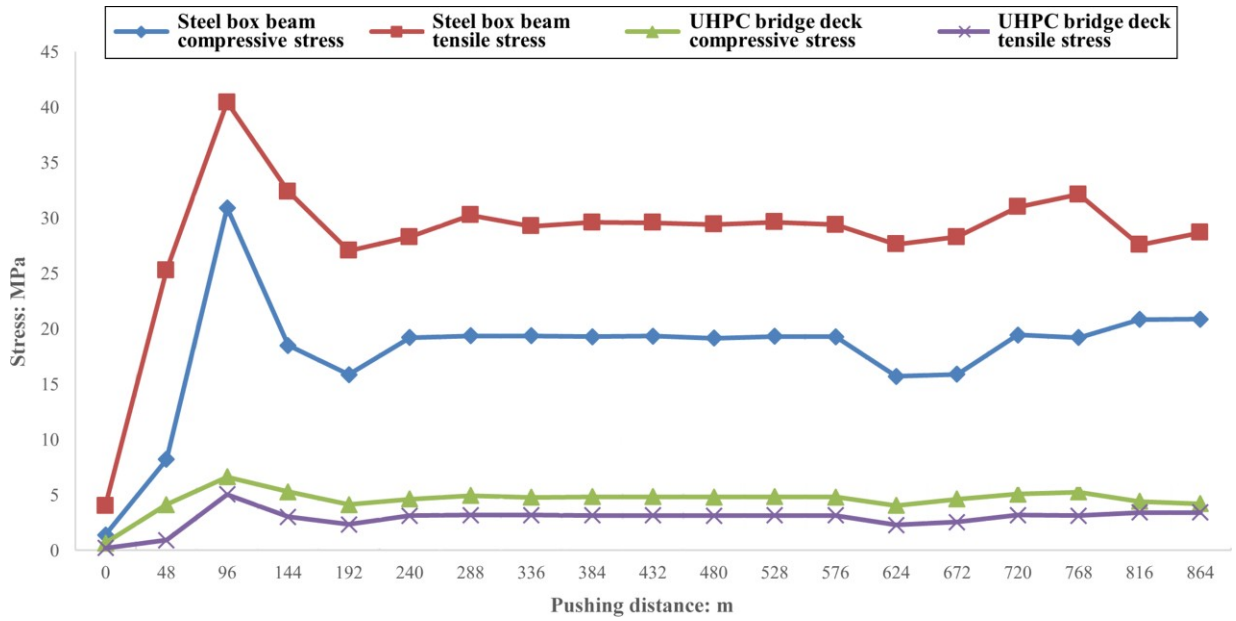


Figure 20. Variation of maximum stress in main beam with pushing distance

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Table 4. Calculation results of main beam stress during the incremental launching construction process

		Maximum box beam stress (MPa)		Maximum stress of UHPC bridge deck (MPa)	
Number	**Distance**	Compressive stress	Tensile stress	Compressive stress	Tensile stress
1	0 m	1.41	4.04	0.67	0.23
2	48 m	8.21	25.29	4.15	0.93
3	96 m	30.92	40.43	6.63	5.07
4	144 m	18.50	32.38	5.31	3.03
5	192 m	15.88	27.04	4.14	2.34
6	240 m	19.22	28.29	4.64	3.15
7	288 m	19.37	30.25	4.96	3.18
8	336 m	19.36	29.26	4.80	3.18
9	384 m	19.29	29.61	4.86	3.16
10	432 m	19.34	29.57	4.85	3.17
11	480 m	19.15	29.41	4.82	3.14
12	528 m	19.32	29.62	4.86	3.17
13	576 m	19.28	29.40	4.82	3.16
14	624 m	15.70	27.64	4.07	2.31
15	672 m	15.89	28.28	4.64	2.56
16	720 m	19.46	30.99	5.08	3.19
17	768 m	19.20	32.12	5.27	3.15
18	816 m	20.84	27.58	4.42	3.42
19	864 m	20.86	28.68	4.22	3.42

intelligence of management. Through the development and implementation of a digital twin information fusion system for construction management, this study supports the information integration and intelligent development in the bridge industry, ensuring the safety and longevity of bridge structures. The main contributions of this study are summarized as follows:

1. This study proposes a logical framework for bridge construction data monitoring and safety assessment based on digital twin technology. At its core, it establishes a dynamic, multi-resolution digital twin model for the incremental launch construction of a steel-concrete composite cable-stayed bridge, integrating the bridge's digital model, structural finite element model, and structural safety evaluation and early warning mechanism.
2. The study establishes a structural safety evaluation and early warning mechanism for the bridge launching process, providing numerical evaluations and early warning plans for critical stress and deformation indicators during bridge construction.

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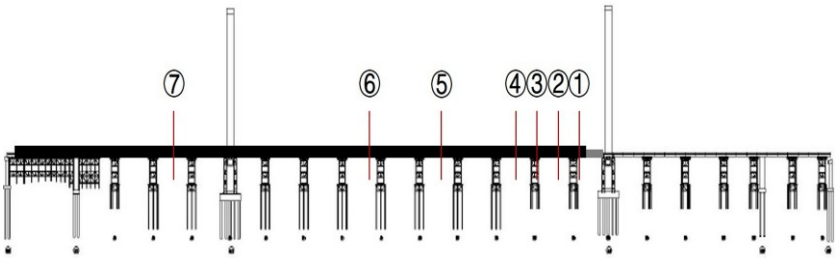


Figure 21. Cross-sectional diagram of stress measurement point arrangement at 573 m pushing distance

Table 5. List of cross-section locations of stress measurement point arrangement at 573 m pushing distance

Section number						
Distance from the front end of the main beam (m)	10	32.5	55	80	170	260	481

Table 6. Comparison of measured and theoretical stress values of main beam at 573 m pushing distance

	Steel box beam stress (MPa)							UHPC bridge deck stress (MPa)				
Section number	Theoretical value	Measured value				Error	Warning level	Theoretical value	Measured value		Error	Warning level
	−10.03	0	0	0.03	0	/	/	1.64	0	0	/	/
	12.41	15.91	12.38	12.36	15.15	11.04%	L1	−1.83	−1.31	−5.29	20.40%	L2
	11.71	13.1	14.82	8.33	14.16	7.62%	No	−1.87	−6.63	5.25	29.70%	L2

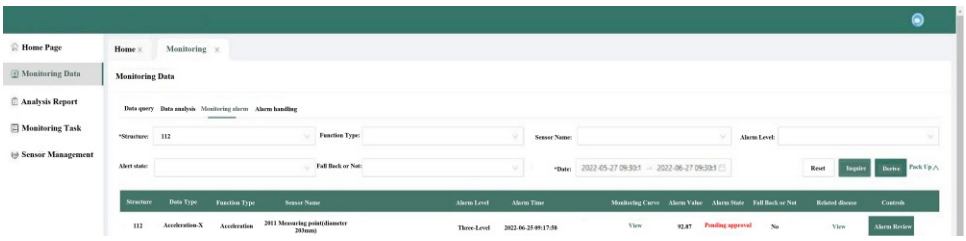


Figure 22. Monitoring and warning module in bridge digital twin system

3. This research integrates the bridge's digital twin model, synchronized structural analysis model, real-time field monitoring data, structural safety evaluation methods, safety early warning mechanisms, emergency response protocols, and field feedback databases to construct a "Bridge Construction Digital Twin Monitoring Platform." By establishing an online digital twin control system, the platform is connected with system platforms, visualization models, and field mobile terminals, enabling efficient control of bridge construction.

This study has some limitations that need to be addressed in future work. First, regarding the validation of theoretical solutions for bridge simulation results, due to limitations in testing conditions, this study did not conduct thorough validation tests. Second, in terms of verifying the effectiveness of the case study, future research will include a comparison between conventional construction safety control methods and the digital twin-based approach proposed in this study. In future research, we will focus on enhancing and validating the functionality and effectiveness of the case study platform based on the proposed research framework. In addition, it will be crucial to enhance the functionality components and improve the applicability of the solution in this study based on the specific characteristics of the project. For example, this could involve establishing a large-scale cable-stayed bridge construction monitoring database to improve the accuracy of safety assessments, as well as integrating dynamic simulation databases of key construction technologies with information databases for bridge maintenance and subsequent major repairs.

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Author contributions

Conceptualization, W.S. and B.Z.; methodology, W.S.; software, F.S.; validation, W.S., F.S.; formal analysis, W.S.; investigation, W.S.; resources, J.C.; data curation, W.S.; writing – original draft preparation, W.S.; writing – review and editing, F.S.; visualization, J.C.; supervision, F.S. and J.C.; project administration, Y.H. All authors have read and agreed to the published version of the manuscript.

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Data availability statement

Please contact the corresponding author to request access to the data mentioned in the article, but note that it cannot be used for commercial activities.

Conflicts of interest

The authors declare no conflicts of interest.

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